

Rooftop Solar Potential Mapping of IIT Delhi Campus: Ten-Week Workflow

Week 1 (18 to 24 Aug). Decide exactly which buildings.

Walk or cycle around campus and list flat-roofed buildings that clearly have no panels. Pick about eight. Also pick two that already have panels, because later you'll use them to check whether your calculations are right. Install QGIS and Python on both your laptops. Send the email asking the Estate Office for the campus electrical drawings and electricity bills. Send it this week because the reply will take a month.

Week 2 (25 to 31 Aug). Collect the raw material.

In QGIS, turn on the satellite image layer so you can see campus from above. Download the building outlines from OpenStreetMap. Most will be roughly right, some will be wrong or missing, and that's normal. Also download the sunlight data for Delhi from the PVGIS website. It gives you an hour-by-hour record of how much sun falls on one square metre for a whole typical year. Save everything in one folder with clear names.

Week 3 (1 to 7 Sep). Draw the roofs properly.

This is boring but it decides everything after it. Zoom into each building in QGIS and trace its roof outline by hand. Then trace every object sitting on that roof: water tanks, staircase rooms, air conditioning units, solar water heaters, lift rooms. QGIS will tell you the area of each shape you draw. Also note how many floors each building has, so you know its height.

Week 4 (8 to 14 Sep). Work out how much roof you can actually use.

You never cover a full roof with panels. You leave about one metre free near the edge, and you leave gaps between rows of panels so one row doesn't shadow the next. So take the roof area, subtract the objects, subtract the edge strip, and then keep only around 40 to 50 percent of what's left for the actual panels. Do the maths for that 40 to 50 percent yourself using the sun's height in winter, don't just copy the number. Put it all in one spreadsheet, one row per building.

Week 5 (15 to 21 Sep). Try the AI part.

Now do the same job automatically. Download a free dataset of aerial photos where buildings are already marked, train a standard image model on it, then run it on your campus images and see if it finds the same roofs you drew by hand. Compare the two. If it matches well, great. If it doesn't, that's still a result, you just report where it failed and why. Your hand-drawn version stays the official one either way. This is your mid-point, so show Prof. Yashasvi your building list and areas.

Week 6 (22 to 28 Sep). Handle shadows.

Tall buildings and trees throw shadows on nearby roofs, and a shaded panel makes far less power. Give every building a height in QGIS so the flat map becomes a 3D-ish model. Then use the UMEP add-on in QGIS, which takes that model and tells you how much sun each roof actually gets across the year after shadows. Check it once against a real shadow you can see in a dated satellite photo.

Week 7 (29 Sep to 5 Oct). Calculate the electricity.

Now you have usable area and real sunlight per roof. Use the Python library pvlib to turn that into units of electricity. It handles panel angle, and it also handles the fact that panels get hot in Delhi summer and lose efficiency. Then subtract the losses: dust on panels, wiring, mismatch between panels, and downtime. Roughly 15 percent in total, plus your shading.

Week 8 (6 to 12 Oct). Check your numbers are believable.

Run your exact method on the two buildings that already have panels, and compare your answer with what those systems really produce. Also check that every building comes out at roughly 1400 to 1550

units per kilowatt per year, which is normal for Delhi. If something is way off, find the mistake now, not in November. Then make the campus map coloured by how good each roof is. Show this to your professor.

Week 9 (13 to 19 Oct). Build the electrical model.

Take one feeder, meaning one 11 kV line that supplies a group of your buildings. Build it in free software called OpenDSS. Put in the transformers, the cable lengths, and each building's electricity usage across the day. If Estate Office never sent data, build a reasonable made-up version and clearly say so in the report.

Week 10 (20 to 26 Oct). See what breaks.

Run the model for a full year, first without solar and then with it. Look for three things. Does the voltage go too high when the sun is strong. Does power start flowing backwards into the transformer, which happens on a summer Sunday when hostels are empty and generation is peak. And how much solar can you add before something crosses a legal limit. That last number is the most interesting result in your project.

After that.

Spend the following week writing up: ranked list of best buildings, total capacity in MW, cost and payback, and clean code that anyone can rerun.

Two rules that matter more than the schedule. Keep one spreadsheet where every building is a row and every number you calculate is a new column, so you can always trace where a figure came from. And put your code on GitHub from Week 1, committing every week, because "we lost the file" is a real thing that happens in October.